Protocol for using HeartbeatDectector.m in MATLAB

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Introduction

This script was designed to process ECG data from animal-borne tags: detecting heartbeats (R waves) semi-automatically, reviewing and correcting them manually, and producing a table of beat times, RR intervals and instantaneous heart rates. It was built so that data are easy both to process and to review and edit afterwards, whether by the original processor or by someone else.

The script is free to use but runs in MATLAB, which requires a license, plus MATLAB's Signal Processing Toolbox. It was tested in MATLAB R2026a (Update 5). At a minimum, the script needs a plain-text data file (e.g. .txt or .csv) containing a column of ECG values, plus either a time column (in seconds) or a known sampling rate. Depth and acceleration can also be included and are displayed alongside the ECG.

In brief, the script: (1) imports the data; (2) crops it to the period the tag was on the animal; (3) filters the ECG; (4) automatically detects candidate heartbeats; and (5) opens an interactive window to review and edit them, saving the results and the settings used.

This protocol has two parts. Part 1 covers the initial processing: setting up the folder structure, running the script, and checking the outputs. Part 2 covers reviewing and editing processed data, either your own or data processed by others.

The script has been tested on deployments from several species, but there is always room for improvement. **If you have any questions or would like to discuss future updates, please contact me at ncnemeth@hawaii.edu.**

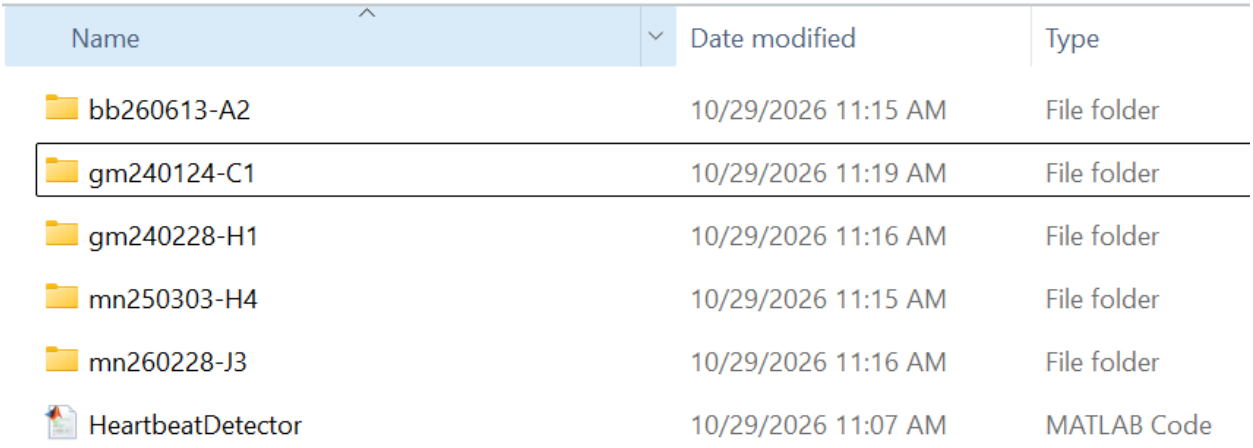
This script and protocol will be soon available on Github, but for now please contact me for how to cite the script.

This protocol was written for Version 3.1 of the script.

Using the script for initial data processing

File structure

Before opening the script for the first time, we need to make sure that your file structure is set up in the way that the script expects. You should have an uppermost level folder that contains all of your deployment folders and the HeartbeatDetector script. In my case, I named this folder HeartRate, but anything will work. So inside of HeartRate (in my case) I would have one folder for each deployment, which looks something like this.



Inside of each deployment folder, at a minimum, needs a folder named "**Data**", with a subfolder "**Raw**", and inside of the "**Raw**" folder is the text file containing your raw data to be processed. The script will make a "**Processed**" folder automatically.

The path to your raw data should look like this:
"C:\Users\Cam\Desktop\HeartRate\gm240124-C1\Data\Raw\gm240124-C1.txt"
You can name the text file whatever you want. I name it the same as the deployment for consistency.

Note the uppermost level folder, "HeartRate" in our case, can live wherever you would like

Opening the script for the first time

Once your file structure is set, you can open the HeartbeatDetector.m script

You should see an introduction at the top, which looks like this

```
%% HeartbeatDetector.m
% Semi-automated R-wave detection and interactive review for ECG tag data
%
% Version 3.1
%
% Created by: Cameron Nemeth, with Claude
% Email: ncnemeth@hawaii.edu
%
% Last modified: 28-10-2026
%
% HOW TO USE
%   1. Keep this script directly in the top-level project folder (e.g.
%      D:\HeartRate): the folder that contains the deployment folders. Not
%      in a subfolder, and not inside a deployment folder. The project
%      folder can have any name and be anywhere.
```

Move to settings by **clicking on the run and advance button** on the "Section" tab on the upper banner, or just scroll down.

Settings section

Start by entering the species of the deployment you are working on, the deployment folder name, and the text file (with the .txt or .csv extension).

```
%% ---- SETTINGS (edit this section for each deployment) ---------------
clear; clc; close all;

settings.species    = 'Pilot whale';          % Species. This is recorded in the settings sheets
settings.deployment = 'gm240124-C1';      % Folder name inside project
settings.rawFile    = 'gm240124-C1.txt';  % raw data text export

% Tag on/off times, in seconds from the start of the recording.
% If known, enter them here. If left empty [], you will click them
% on the overview plot.
settings.tagOnTime  = [];
settings.tagOffTime = [];
```

You may leave tag on/off times blank for now, or enter the time in seconds if known. There will be a step to determine these from the data later, if depth and/or acceleration data are present.

Next, enter the column names or numbers for your various datastreams (ECG, depth, etc.) so the script knows where to find the data

```
% Columns to use. Give each one EITHER as a NAME in quotes, exactly as it
% is written in the file, OR as a column NUMBER, counting the columns of
% numbers from the left (1 = first column).
%   LabChart export: use names from the ChannelTitle line. The first
%     column is time, which LabChart leaves unnamed: use 'Time' (or 1).
%     Careful with numbers here: because time comes first, the first name
%     on the ChannelTitle line is column 2, not column 1.
%   File with a row of column names (e.g. Time,ECG,Depth): use the names.
%   File of numbers only: use column numbers.
% The script prints which columns it used, so check that after importing.
% Depth is optional: if the file has no depth, set depthChannel = []. The
% depth strips in the windows are then left empty; nothing else changes.
settings.timeChannel  = 1;        % time column (seconds), or [] if the file has none
settings.ecgChannel   = 'ECG';         % raw ECG values
settings.depthChannel = 'Depth (m)';   % depth values, or [] if the file has none

% Acceleration (optional): one setting per axis, each a column name or
%   number, or [] if that axis is not in the file. Fill in all three for a
%   3-axis tag, just one for a single-axis tag, or leave all three [] for
%   none. Acceleration is for display only (a strip above depth, to help
%   spot movement, e.g. muscle noise in the ECG); it is not used for
%   detection.
settings.accelX = 'Acceleration 1';                % X axis
settings.accelY = 'Accleration 2';                 % Y axis
settings.accelZ = 'Accleration 3';                 % Z axis
```

These can be found in your raw text file.

```
Interval=       0.004 s
ExcelDateTime=  2.5569000000000000e+004  0:00:00
TimeFormat=     StartOfBlock
DateFormat=
ChannelTitle=   ECG     Depth (m)       Acceleration 1  Accleration 2   Accleration 3
Range=  10.000 V        10.000 V        10.000 V        10.000 V        10.000 V
0       770     -900    -900    -900    -900    -900    -0.2980674      0.2980676
0.004   770     -900    -900    -900    -900    -900    -0.2980674      0.2980676
0.008   770     -900    -900    -900    -900    -900    -0.2980674      0.2980676
0.012   770     -900    -900    -900    -900    -900    -0.2980674      0.2980676
0.016   770     -900    -900    -900    -900    -900    -0.2980674      0.2980676
```

Note: Column names go in ‘’, column numbers stand alone (see settings.timeChannel above), and if not applicable place empty brackets [].

The sampling rate setting is **NOT** necessary and should be left blank unless you do not have a time column. The script uses the interval between times to calculate the sampling rate.

```
% Sampling rate (Hz): ONLY needed if the file has no time column
%   (settings.timeChannel = []). Times are then counted from 0 s at the
%   first row. If the file does have a time column, the rate is worked out
%   from it and this value is ignored (a note is printed). Example: 250.
settings.sampleRateHz = [];

% Notes for this deployment, saved in the settings sheets (e.g. 'ECG shifted
% 17 s in LabChart to align with CATS depth'). Leave as '' if none.
settings.notes = '';
```

Any deployment level notes can be left in settings.notes

Run and advance when done.

Paths

You can run and advance through this section. No settings need to be tweaked.

```
%% ---- PATHS (no editing needed) ------------------------------------
% Find the project folder (MATLAB's version of R's here()): the folder that
% contains this script and the deployment folders. Some MATLAB versions run
% a temporary copy of the script (e.g. when there are unsaved changes), so a
% few likely places are checked, and the first one that contains the
% deployment folder is used.
candidates = {fileparts(mfilename('fullpath'))};            % where the script is running from
try
    candidates{end + 1} = fileparts(matlab.desktop.editor.getActiveFilename);   % file open in the Editor
catch
end
candidates{end + 1} = pwd;                                  % MATLAB's current folder
```

Toolbox check

You can run and advance through this section. No settings need to be tweaked.

```
%% ---- TOOLBOX CHECK ------------------------------------------------
% Filtering and peak detection later on need the Signal Processing Toolbox.
if isempty(ver('signal'))
    warning(['Signal Processing Toolbox is not installed. It will be needed ' ...
        'for filtering and peak detection. Install via Home > Add-Ons.']);
else
    fprintf('Signal Processing Toolbox: found\n');
end
```

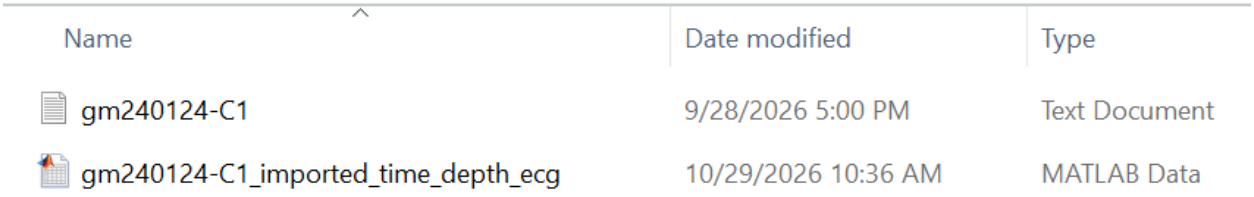
Import

You can run and advance through this section. No settings need to be tweaked.

This step will read in the information from the raw text file and saves a .mat file of just the columns used by the rest of the script. Future runs load the .mat file instead to speed up loading times in the case of large datasets.

```
%% ---- IMPORT ----------------------------------------------------
% Reads time, ECG and depth from the raw text file. Three kinds of file are
% recognized automatically:
%   1. LabChart "Export As Text" with the block header: column names come
%      from the ChannelTitle line, and the first (unnamed) column is time,
%      which the script calls 'Time'.
%   2. A file with one row of column names above the numbers
%      (e.g. Time,ECG,Depth). If there are several lines above the numbers,
%      the last one is taken as the column names.
%   3. A file of numbers only (no column names).
```

The .mat file will look something like “gm240124-C1_imported_time_depth_ecg” and will be in the raw data folder.



The command window should print some diagnostic information including: the column names, sampling rate, number of samples, duration, irregularities in the sampling intervals, na values, etc.

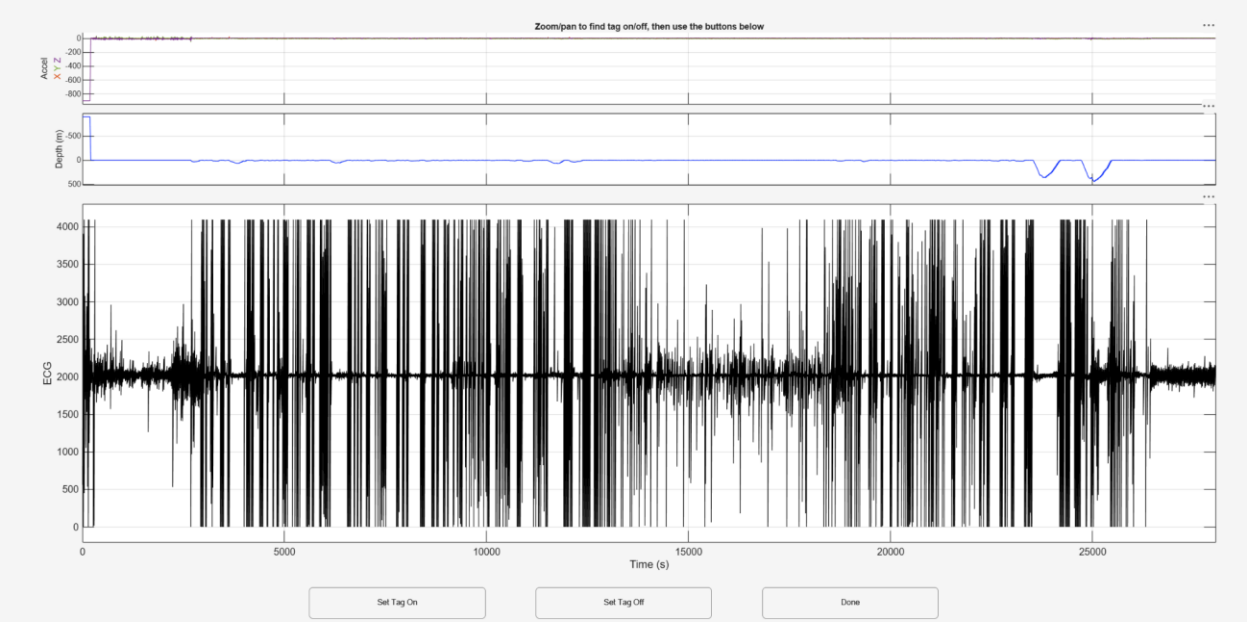
```
Loaded saved import: D:\HeartRate\gm240124-C1\Data\Raw\gm240124-C1_imported_time_depth_ecg.mat
File type: LabChart export (columns separated by tabs)
Columns used: Time = column 1 (Time)  |  ECG = column 2 (ECG)  |  Depth = column 3 (Depth (m))
Acceleration: Acc X = column 4 (Acceleration 1)  |  Acc Y = column 5 (Accleration 2)  |  Acc Z = column 6 (Accleration 3)
Sampling rate: 250 Hz (from the time column)

Samples: 7011482 | Sampling rate: 250 Hz | Duration: 7.79 h
Irregular time steps: 0
NaN values - ECG: 0, Depth: 0
```

Overview plot and tag on/off

There are no settings to be adjusted here, so you can run the section.

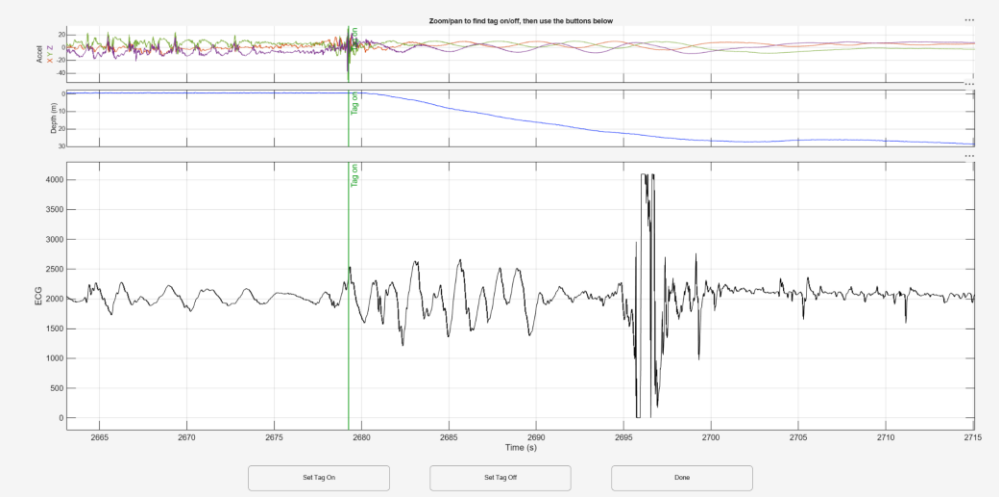
If you did not enter tag on/off times earlier, you should see something like this, with the whole deployment visible and buttons at the bottom to select the tag on and tag off times.



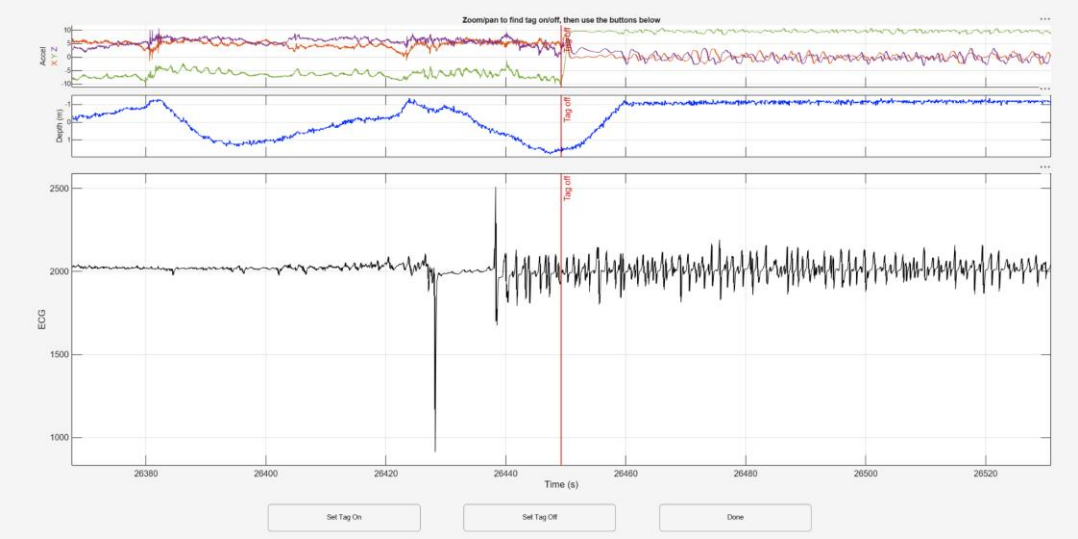
Scroll to zoom, drag the panel to move left and right

Scroll through the data, find where the tag goes on the animal, click "Set Tag On". If unknown, just choose the very beginning to be safe. You will just need to go through more of the data. This step just crops the data into the periods where the tag is on the animal.

Tag on image



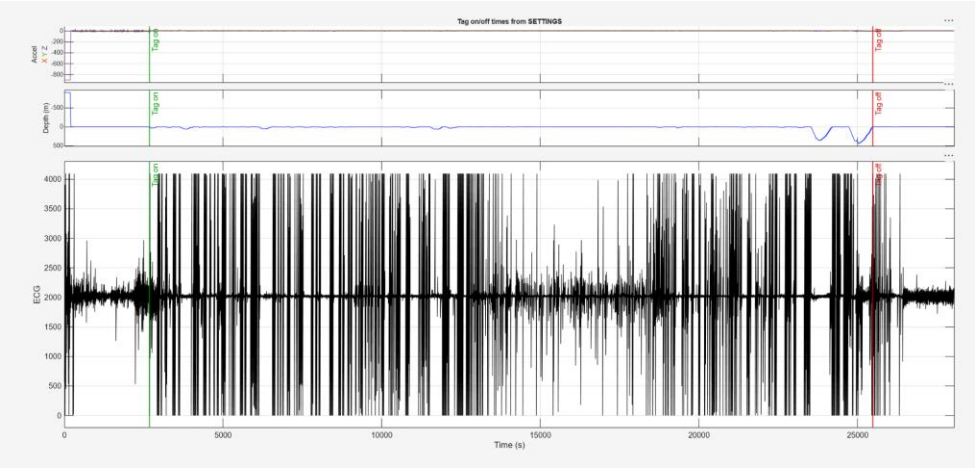
Tag off image



A popup box will appear with the option to accept these times or to redo. To redo, close the box and run the section again. The console will also show this message. More on the settings sheet later.

```
Tag times accepted and used for this run:
  settings.tagOnTime  = 2679.440;
  settings.tagOffTime = 24973.160;
They will be saved in the settings sheets when the review window opens.
Optional: paste the two lines above into SETTINGS (now or after the run)
so later runs use the same times without clicking.
Cropped to 2679.440 - 24973.160 s: 5573431 samples (6.19 h)
```

If you had already entered tag on/off times at the very beginning settings, you should see something like this, where the green line is the tag on, and the red line is the tag off.



Filtering

A band-pass filter is applied to the raw data using the settings specified.

```
%% ---- FILTERING ----------------------------------------------------------
% Makes a filtered copy of the ECG, used only for beat detection.
% The raw ECG (data.ECG) is never changed.
% Band-pass filter from settings.highPassHz to settings.lowPassHz, run
% forward and then backward (filtfilt) so filtered peaks stay at the same
% times as the raw peaks.


% ----- FILTER SETTINGS -----
% Values used are recorded automatically in <deployment>_Settings.csv and
% HeartRate_Reference_Settings.csv (see the note at the top of the script).
%
%
% High-pass cutoff (Hz): removes slow drift (baseline wander from breathing,
%   body movement, electrode shifts, depth changes). Raising it flattens the
%   baseline more, but too high can shrink wide/slow QRS complexes.
%   Typical: 0.5-5 Hz.
settings.highPassHz = 1;

% Low-pass cutoff (Hz): removes fast jitter. Lowering it smooths more, but
%   too low can blunt sharp R peaks. Must be below half the sampling rate
%   of the tag. Typical: 25-50 Hz.
settings.lowPassHz = 40;

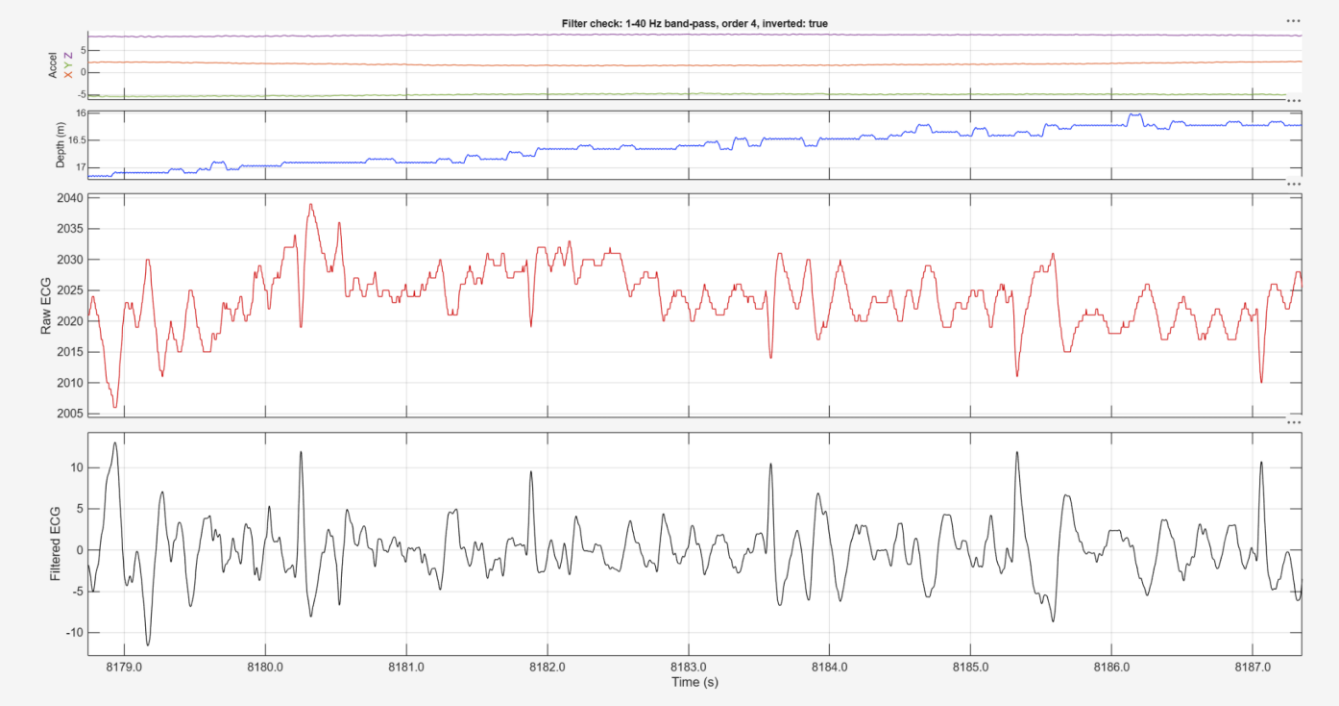
% Filter order: how sharply the filter cuts off around the two cutoffs.
%   Higher = sharper, but can add small ripples ("ringing") next to large
%   peaks. 2-4 is typical; leave at 4 unless there is a reason to change it.
settings.filterOrder = 4;

% Invert ECG: set to true if R waves point DOWNWARD on this tag. The script
%   prints a suggestion each run. Only the filtered copy used for detection
%   is flipped; the raw ECG is never changed.
settings.invertECG = true;
```

This setting below can always stay true

```
% THIS CAN JUST STAY TRUE
% False just skips the figure pane for this filtering step.
% Show the filter check plot (raw and filtered ECG, below any acceleration
%   and depth strips). Set to false once the values above are settled, to
%   skip it.
settings.showFilterCheck = true;
```

Try different settings, view the pop up window, and rerun with different settings if needed.



Automatic R-Wave detection

```
%% ---- AUTOMATIC R-WAVE DETECTION ----------------------------------------
% Finds every upward peak in the filtered ECG, measures how much each one
% stands out from the nearby background (size, steepness, width), and keeps
% those that look like R waves. Results are shown as red circles for a quick
% check. Nothing is saved here; beats are locked in during the review step.
```

Edit the below settings to automatically detect peaks in the signal.

Note: Different deployments are inherently more noisy. This works better on some than others and at different times within the same deployment. The goal here is not perfection, but the best we can do. The manual review makes it easy to remove incorrect points.

```
% ----- DETECTION SETTINGS (adjust here while tuning) -----
% Stands-out ratio (AMPLITUDE): how many times taller a peak must be than
%   the typical background signal nearby. Raise it if noise bumps get red
%   circles; lower it if small real beats are missed. Typical: 3-8.
settings.minStandOut = 4;

% Steepness ratio (SLOPE): how many times steeper a peak's rise or fall must
%   be than the typical background nearby. R waves are sharp; slower bumps
%   (T waves, movement) are not. Raise it to reject rounded bumps; lower it
%   if real beats are missed. Typical: 3-8.
settings.minSteepness = 4;

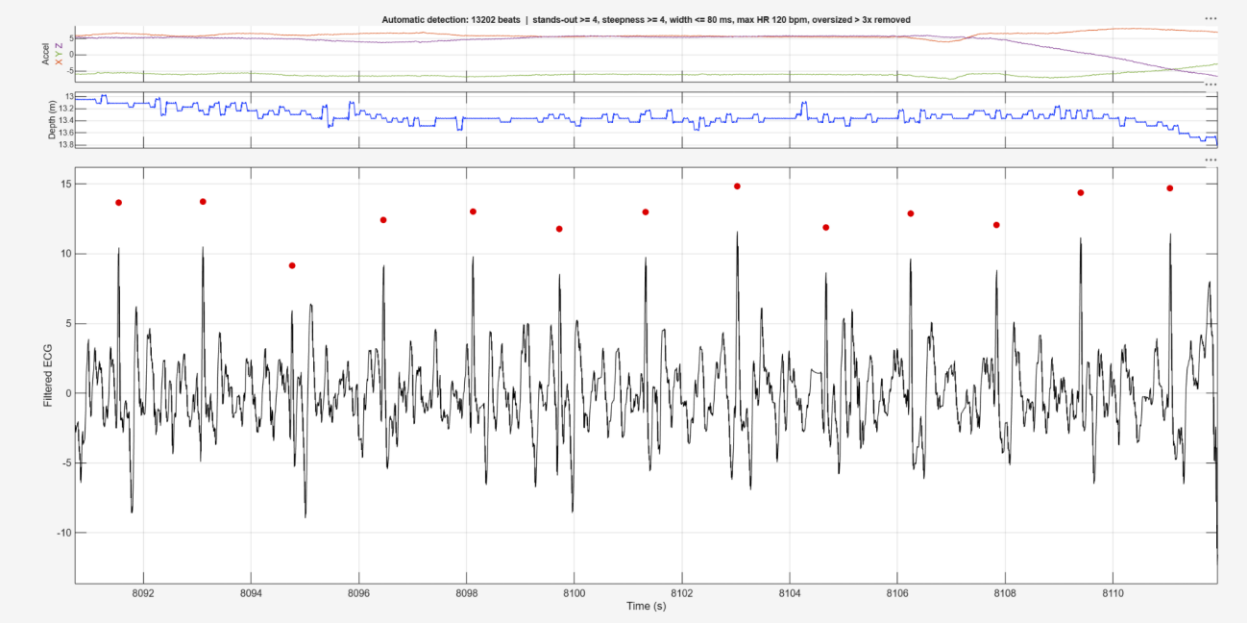
% Maximum width (ms): the widest a peak can be and still count as an R wave,
%   measured at half its height above zero (the filtered ECG is centred on
%   zero, so this measures the sharp tip of the R wave). Lower it to reject
%   broad bumps; raise it if real beats are missed for being too wide.
%   Typical: 50-120 ms.
settings.maxWidthMs = 80;

% Noise window (s): how many seconds around each peak are used to judge the
%   "typical background". Shorter adapts faster to noisy patches; longer is
%   steadier. Typical: 3-10 s.
settings.noiseWindowS = 5;

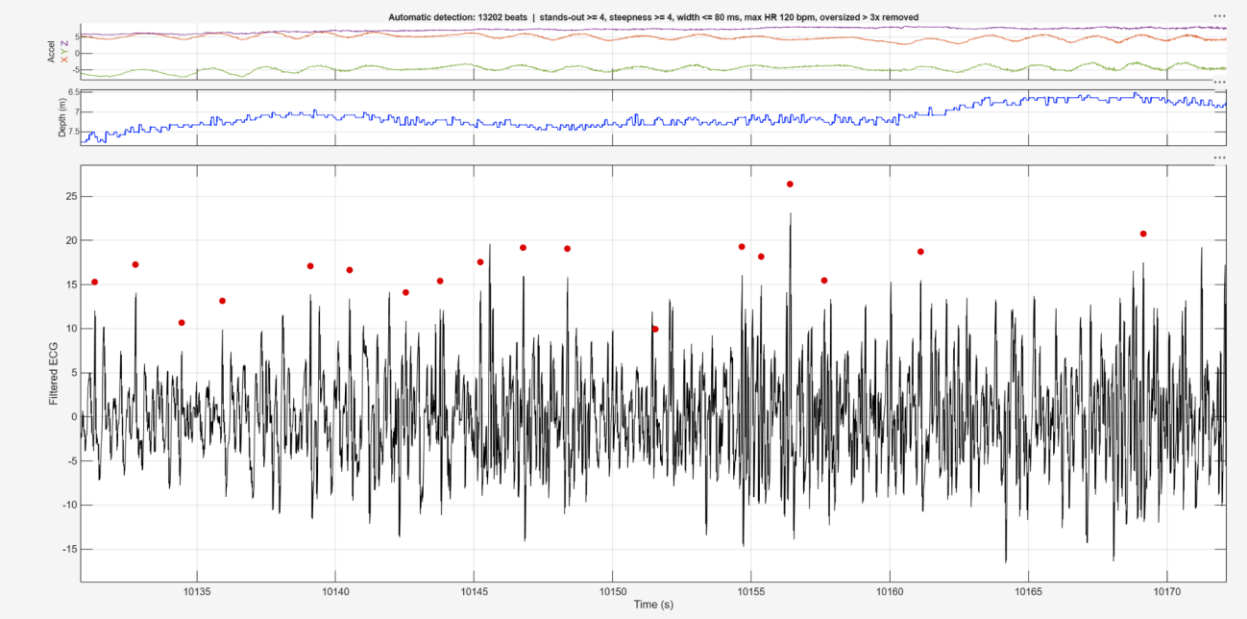
% Maximum heart rate (bpm): two beats can never be closer together than
%   this allows (120 bpm = 0.5 s). If two peaks are closer, only the one
%   that stands out most is kept.
settings.maxHR = 120;


% Oversized-peak filter (removes e.g. surfacing artifacts): a detected beat
%   is removed if its prominence is more than maxRelSize times the median
%   prominence of its neighbouring beats (nNeighbors on each side).
%   Lower maxRelSize to remove more large peaks; raise it if real beats are
%   being removed. Typical: 2-5 times, 3-10 neighbours each side.
settings.maxRelSize = 3;
settings.nNeighbors = 5;
```

Some of these automated detections are good in some sections



And not so good in others… Fortunately removing these dots is pretty easy in the next section, so if you can find nice sections of heartbeats that are being detected well, then it is probably best to move to the review stage.



Review: Edit beats and mark breaks

```
%% ---- REVIEW: EDIT BEATS AND MARK BREAKS ---------------------------------
% Edits are saved to <deployment>_Heartbeats.csv in the Processed folder
% when you click Save & keep working or Save & finish (bottom right). Save
% every few minutes: unsaved edits are lost if MATLAB closes unexpectedly.
% The CSV can be open in Excel while you work, but close it there before
% saving, or the save is blocked (a warning says so; your edits are kept).
%
```

This is the main section of this script. This is where you will manually go through the deployment and select beats and remove inaccuracies chosen by the automatic detector in the last step.

Some instructions are given in the header

```
% MARKERS
%   Red circle    = certain beat  (Certainty 1)
%   Green circle  = possible beat (Certainty 2; You are ~80% sure that beat is real)
%   Red x         = certain beat with no known beat before it (NoPriorBeat 1).
%                   Its RRInterval and HeartRate are NaN. The next beat
%                   after an x DOES get an RR interval, measured from the x.
%   The first beat of the deployment is always an x (nothing comes before it).
%
% MOUSE AND KEYS
%   Click on empty space        -> add a red circle
%   Shift + click               -> add a green circle
%   Ctrl + click                -> add a red x
%   Click on any marker         -> remove it (any color or shape)
%   Drag / scroll               -> move / zoom in time
%   Shift + drag                -> select a time span (shaded blue band)
%   Delete or Backspace         -> remove every marker in the selected span;
%                                  the first beat AFTER the span becomes an x
%   Esc (or any click)          -> cancel the selection
%   Left / right arrow keys     -> move back / forward by most of a window
%   Z                           -> undo the last edit (repeat to undo more)
%   C (pointer over a marker)   -> add, edit or remove a comment on that beat
%                                  (beats with a comment get a black ring)
%   Save & keep working button  -> save now and carry on (orange = unsaved
%                                  changes, green = everything saved)
%   Save & finish button        -> save, record progress, and close
%   Closing the window (X)      -> asks to save first if there are unsaved changes
%   New markers snap to the highest point of the filtered ECG near the click.
%   NOTE: MATLAB treats a RIGHT click the same as Ctrl + click (adds a red x)
%   and a MIDDLE click the same as Shift + click (adds a green circle).
```

These also appear on the panel that displays the data.

There are a few settings to be tweaked in this section.

Notably, minimum heart rate. This is just an estimate and does not have any true effect on the data, but the script can use this information to automatically detect the first beat of a segment when the prior signal is noisy.

```
% Minimum heart rate (bpm). This is just an approximation.
% For example, 20 bpm = any gap over 3 s gets an x.
% Lower it if real slow beats (e.g. deep dives) get x's
% Raise it to flag shorter gaps.

settings.minHR = 20;

% Break factor: how much longer than the typical nearby gap a gap must be to
%   get an x. Example: beats every 1.5 s, factor 1.5 -> any gap over
%   1.5 x 1.5 = 2.25 s gets an x. One missed beat doubles the gap (3.0 s),
%   so it is caught; normal variation (e.g. 1.4-1.7 s) is not.
%   Lower it to catch more missed beats (but more x's on real slowdowns);
%   raise it for fewer x's on real slowdowns (but some missed beats may be
%   missed). Typical: ~ 1.4-2.
settings.breakFactor = 1.5;

% Break neighbours: how many gaps on each side define the "typical gap".
%   Typical: 3-10.
settings.breakNeighbors = 5;
```

There are also some settings that are personal preferences. These are not recorded by the script and can vary within and between deployments with no effect on the data.

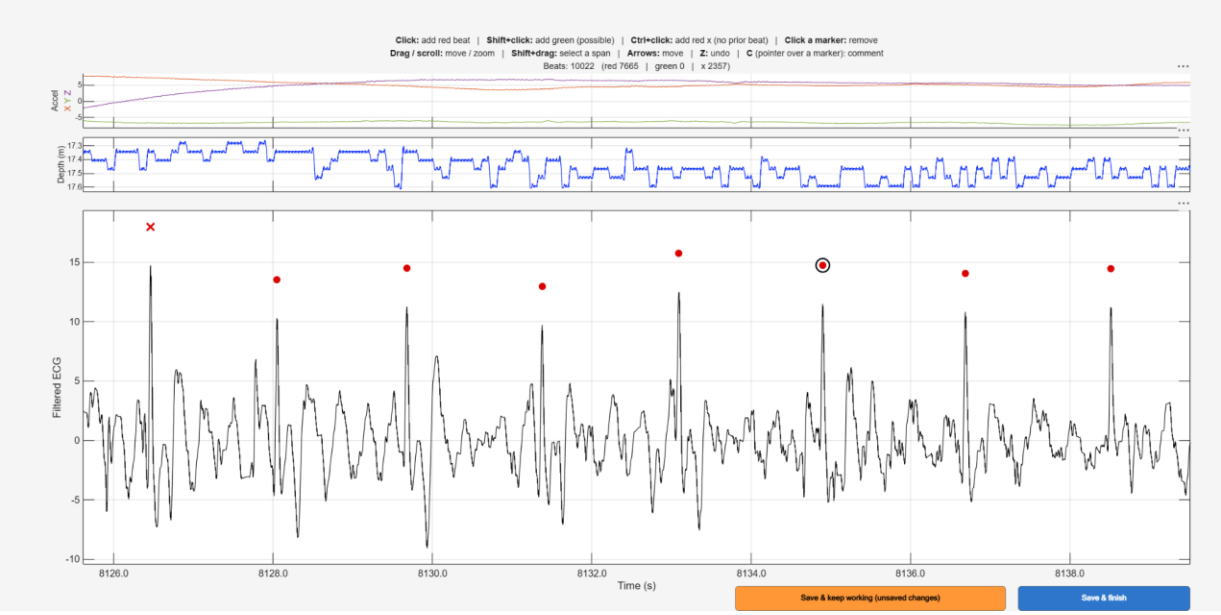
```
% Review window length (s) when the window first opens. Zoom freely after.
% Really not much of an effect here.
settings.reviewWindowS = 20;

% Snap distance (ms): a new marker is placed on the highest point of the
%   filtered ECG within this distance of your click. Raise it if markers land
%   beside the peak; lower it if they jump to a neighboring bump.
%   Typical: 50-150 ms.
settings.snapMs = 100;

% Click tolerance (pixels): how close a click must be to a marker to count
%   as clicking ON it (which removes it). Raise it if removing markers takes
%   precise aim; lower it if clicks near a marker remove it by accident.
%   Typical: 5-12 px.
settings.clickTolerancePx = 8;
```

Once the settings are in place, you can run this section of the script.

This is an example of a section. The red x marks the first beat in a segment of successive beats.



This x will not record an R-R interval for that beat, given that the prior beat is unknown.

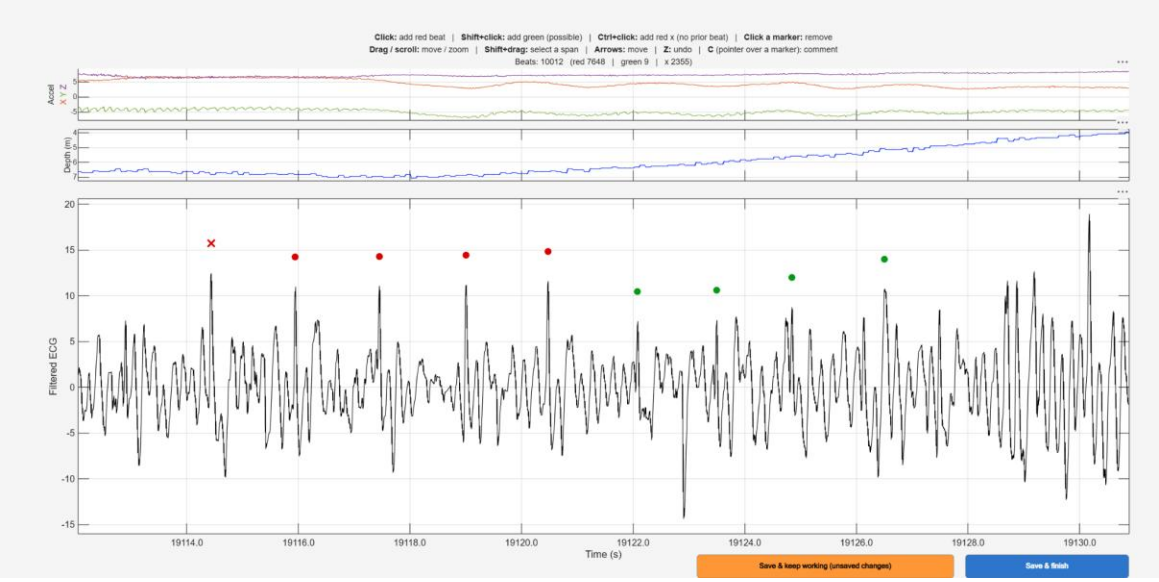
Red dots indicate a beat in which the prior beat is known.

These will contain a R-R interval and instantaneous heart rate

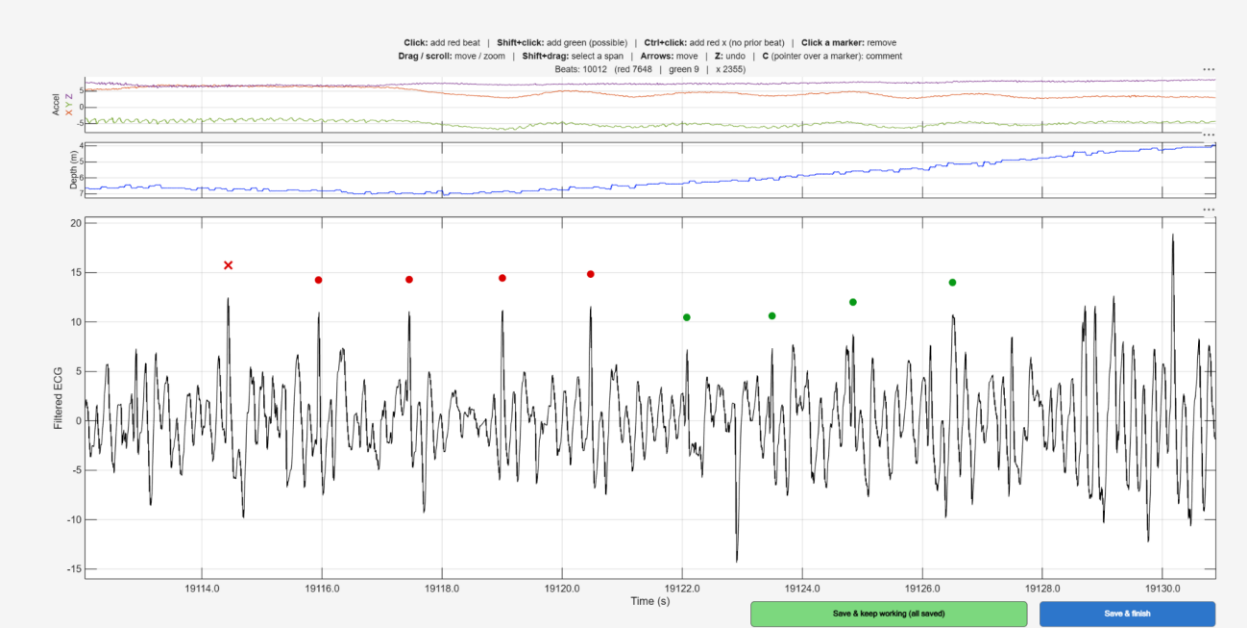
Comments for a specific beat are given by hovering over a beat and clicking “c”.

These appear as a black circle outlining a beat.

Green dots (shown below) indicate a degree of uncertainty. These get a “Certainty” value of “2” in the final datasheet, as opposed to a standard “1”, making them easy to filter out or inspect later

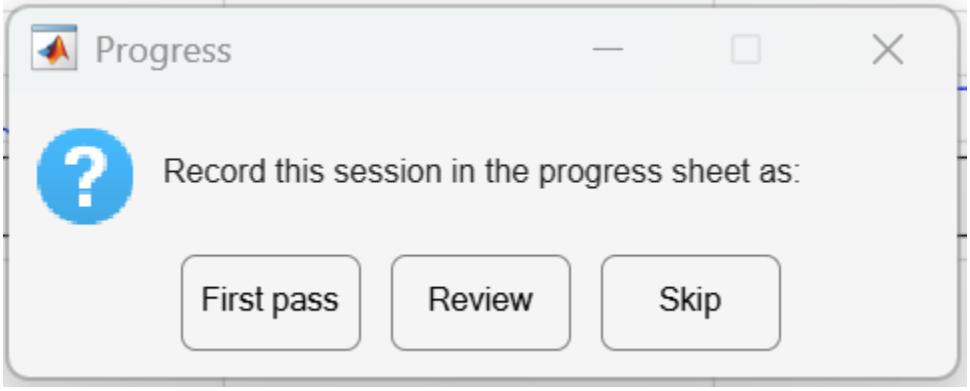


Edits DO NOT save as you go, so you should periodically save your edits using the "Save and keep working" orange button in the bottom right hand corner. This button turns green when all changes are saved.



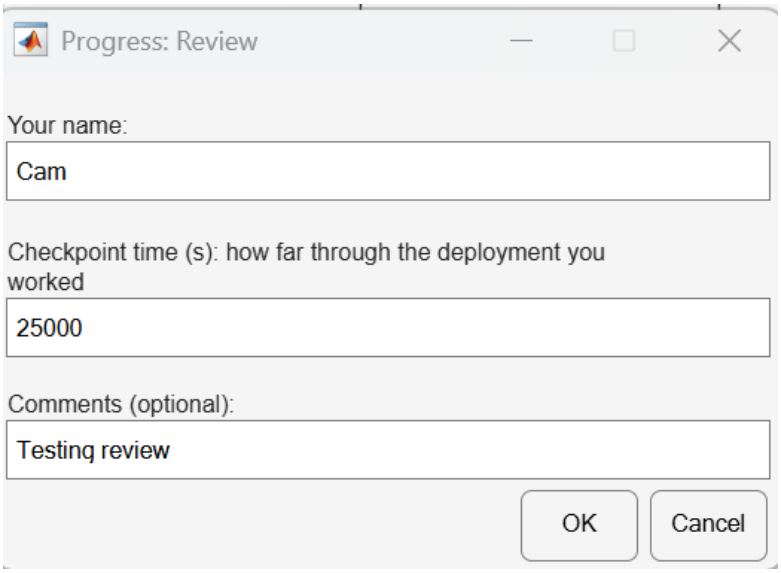
When you are done working for the session, whether you completed only a portion of the deployment or the entire thing, click "**Save & finish**"

A popup box will appear

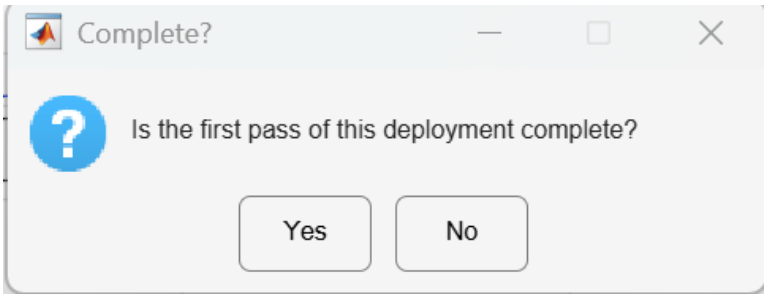


If you are still working to complete the first pass of the deployment (i.e., not a supervisor checking the work of someone else or yourself) click "**First pass**"

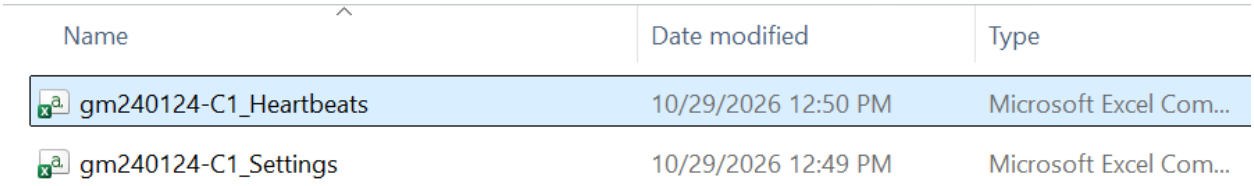
The next popup box will appear. Enter your name, the time in the deployment that you have worked through, and any comments you want to make.



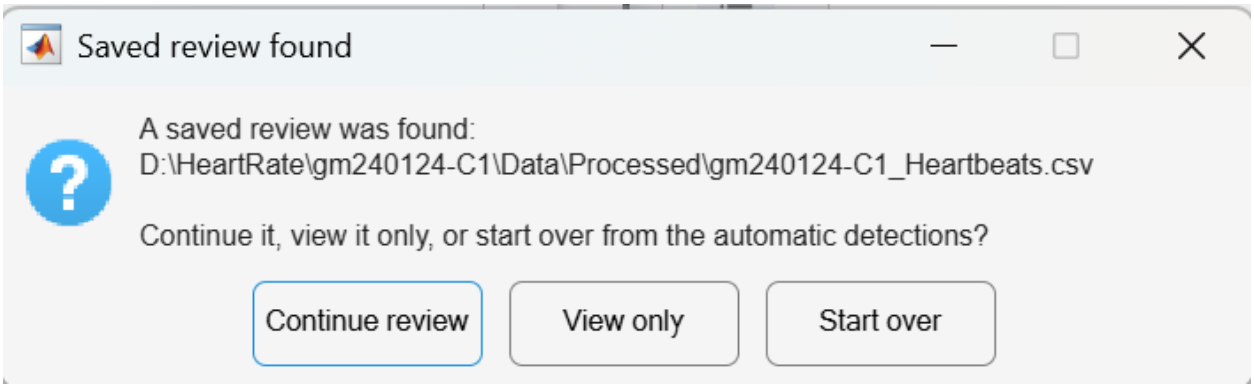
And one final box will appear. Click **"Yes"** if you have completed the deployment and **"No"** if not.



If you are returning to the script later, repeat the process, but you can find your previous settings in the "deployment_Settings" excel sheet. This will allow you to quickly run through everything and get to the review section.



Once you run the Review section, you will get a popup box that looks like this



Continue review: reopens the saved review from "deployment_Heartbeats", with all beats, x's and comments exactly as they were saved, and all editing options available. The window opens at the start of the tag-on period; use the checkpoint time in "HeartRate_Progress" to find where work stopped.

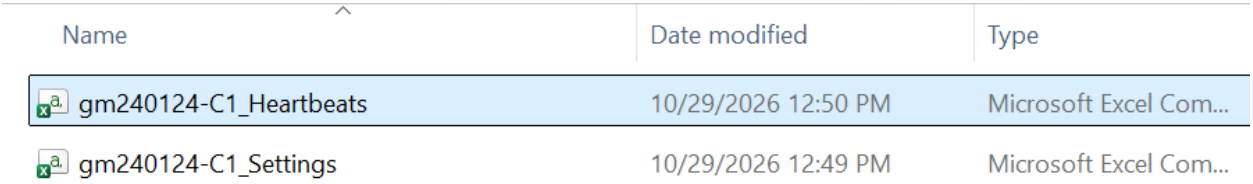
View only: shows the saved review with editing switched off, so nothing can be changed by accident. Useful for checking someone else's work. Close the window with the **Close** button.

Start over: discards the saved review and starts again from the beats found by the automatic detector (with x's pre-placed again). A confirmation popup appears first. The saved file is only replaced when you next save, so closing without saving leaves it unchanged; once saved, all previous edits and comments are lost.

At this point you have completed everything and the script should have saved various outputs, which we will now explore.

Outputs

In your processed data folder, you should see two excel sheets. One of them (deployment_Heartbeats) contains all of the processed data, while the other (deployment_Settings) contains all of the settings used throughout the script. For consistency, these settings can be used later when returning to a deployment.



Your deployment_Hearbeats excel sheet will look something like this

	A	B	C	D	E	F	G
1	Time	Certainty	NoPriorBeat	RRInterval	HeartRate	Segment	Comment
2	2680.976	1	1	NaN	NaN	1	
3	2682.16	1	0	1.184	50.67567568	1	
4	2686.004	1	1	NaN	NaN	2	
5	2697.26	1	1	NaN	NaN	3	
6	2700.212	1	0	2.952	20.32520325	3	
7	2703.9	1	1	NaN	NaN	4	
8	2704.652	1	0	0.752	79.78723404	4	
9	2705.3	1	0	0.648	92.59259259	4	
10	2709.88	1	1	NaN	NaN	5	
11	2710.52	1	0	0.64	93.75	5	
12	2711.16	1	0	0.64	93.75	5	

Time: the time of the R-wave peak for that beat, in seconds from the start of the recording (the same time base as the raw data file, not from tag-on). It is the highest point of the filtered ECG at that beat, to the nearest sample.

Certainty: “1” for all red circles, “2” for any green circles. This allows the user to select beats they are not 100% confident in, whenever that may be necessary, and for any supervisors to easily review that section and/or filter out those data.

NoPriorBeat: The first heartbeats within a segment. There was likely noise before this heartbeat, thus this particular beat does not have a R-R interval or instantaneous heart rate (both columns appear as NaN).

RRInterval: The duration (in seconds) between the previous heartbeat and the current heartbeat.

HeartRate: Instantaneous heart rate of that beat, calculated as 60 divided by the R-R interval
Segment: Segment: a number (1, 2, 3, …) identifying each continuous stretch of beats, meaning a run of beats with no known gaps between them. A new segment starts at every beat with NoPriorBeat = 1 (a red x), and continues until the beat before the next x. The first beat of the deployment always starts Segment 1.

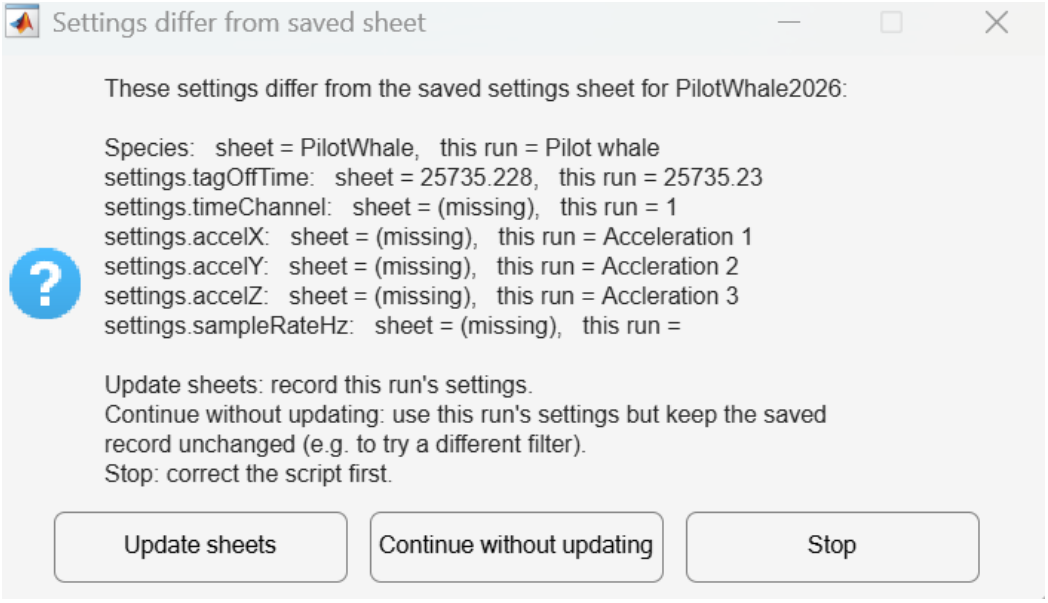
Your deployment_Settings excel sheet will have columns that look something like this

M	N	O	P	Q	R	S	
settings.highPassHz	settings.lowPassHz	settings.filterOrder	settings.invertECG	settings.minStandOut	settings.minSteepness	settings.maxWidthMs	se
1	40	4	TRUE	4	4	80	

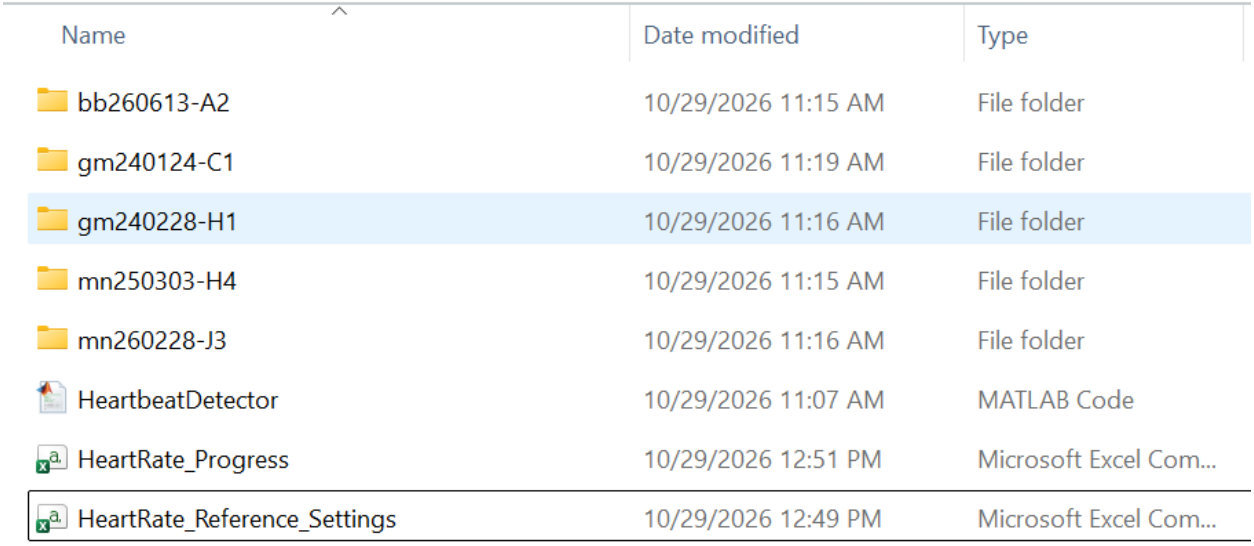
*This is not all of them shown above

These are used for consistency when returning to the script. If you started to manually edit the beats and are now returning to the script, use this sheet for a reference of what was chosen last time.

If you rerun the script with different settings, you will get a popup box that look like this, explaining what is different. You have three options, to update the Settings sheet, to continue without updating the settings sheet, or to stop the run.



There are also two new excel sheets that have been added to your upper-level project folder



Name	Date modified	Type
bb260613-A2	10/29/2026 11:15 AM	File folder
gm240124-C1	10/29/2026 11:19 AM	File folder
gm240228-H1	10/29/2026 11:16 AM	File folder
mn250303-H4	10/29/2026 11:15 AM	File folder
mn260228-J3	10/29/2026 11:16 AM	File folder
HeartbeatDetector	10/29/2026 11:07 AM	MATLAB Code
HeartRate_Progress	10/29/2026 12:51 PM	Microsoft Excel Com...
HeartRate_Reference_Settings	10/29/2026 12:49 PM	Microsoft Excel Com...

HeartRate_Progress (or whatever name of folder you chose instead of HeartRate)

	A	B	C	D	E	F	G	H	I	J
	Deployment	ProcessedBy	ProcessedDate	CheckpointTime	FirstPassComplete	ReviewedBy	ReviewDate	ReviewCheckpointTime	ReviewComplete	Comments
	bb260613-A2	Cam	6/30/2025	Finished	Yes					Weird section at 1000
	gm240124-C1	Cam	6/30/2025		Yes					
	gm240228-H1	Cam	9/4/2025		Yes					
	mn250303-H4	Cam	10/29/2026	2699	No					Deep dives
	mn260228-J3	Cam	10/31/2026	19131	Yes					

This is a running list of your progress across all of your deployments. At this stage, prior to any review, only columns A-E should be populated. These come from those last popup boxes after the manual processing of the data.

HeartRate_Reference_Settings

This is a running list of the settings you used for all deployments. This can be useful for someone who may be unfamiliar with a species and wants to see the settings used.

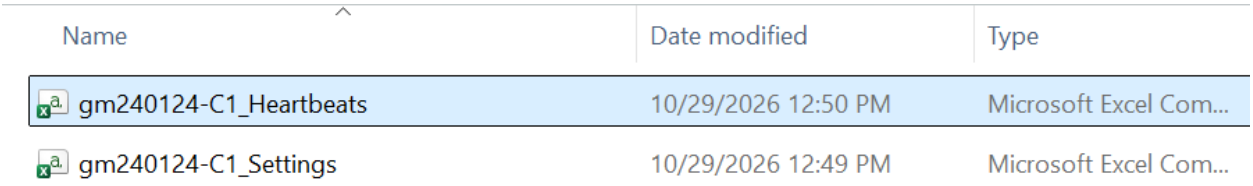
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Species	Deployment	settings.rawFile	settings.tagOnTime	settings.tagOffTime	igs.timeCh	ngs.ecgCh	gs.depthCh	ttings.acc	ttings.acc	ttings.acc	s.sampleFn	gs.highPa	ngs.lowPa	ings.filterO	ings.invert	igs.minSta	s.minSte
	bb260613-A2	bb260613-A2	bb260613-A2.txt	507.74	4189.704	1	2						1	40	4	FALSE	20	3
	gm240124-C1	gm240124-C1	gm240124-C1.txt	1100	17000	1	ECG	Depth (m)					1	40	4	FALSE	20	3
	gm240228-H1	gm240228-H1	gm240228-H1.txt	0.184	23261.64		ecg	depth					1	40	4	TRUE	20	3
	mn250303-H4	mn250303-H4	mn250303-H4.txt	2679.44	24973.16	1	ECG	Depth (m)	celeration	celeration	celeration	3	1	40	4	TRUE	4	4
	mn260228-J3	mn260228-J3	mn260228-J3.txt	2679.668	25735.23	1	ECG	Depth (m)	celeration	celeration	celeration	3	1	40	4	TRUE	4	4

At this point you have now run through the script for the deployment and are familiar with all outputs. The next parts of this protocol are only relevant to anyone reviewing the data or supervising students using the script.

Using the script to review and modify processed data

When reviewing the work of yourself or others, you will move section by section through the script, just as you did the first time.

Use the “deployment_Settings” sheet in the processed data folder to find all of the settings used in the previous run.

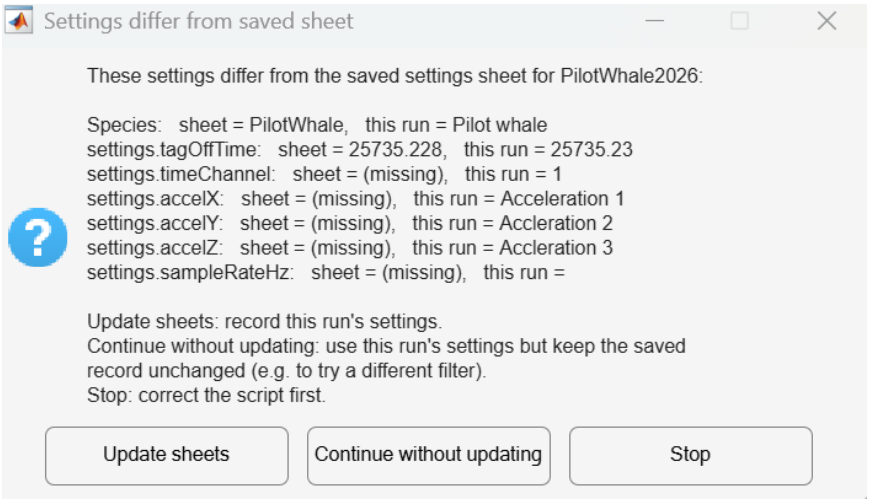


This looks something like

M	N	O	P	Q	R	S	
settings.highPassHz	settings.lowPassHz	settings.filterOrder	settings.invertECG	settings.minStandOut	settings.minSteepness	settings.maxWidthMs	se
1	40	4	TRUE	4	4	80	

This will allow you to easily move through the sections while being consistent with the previous run.

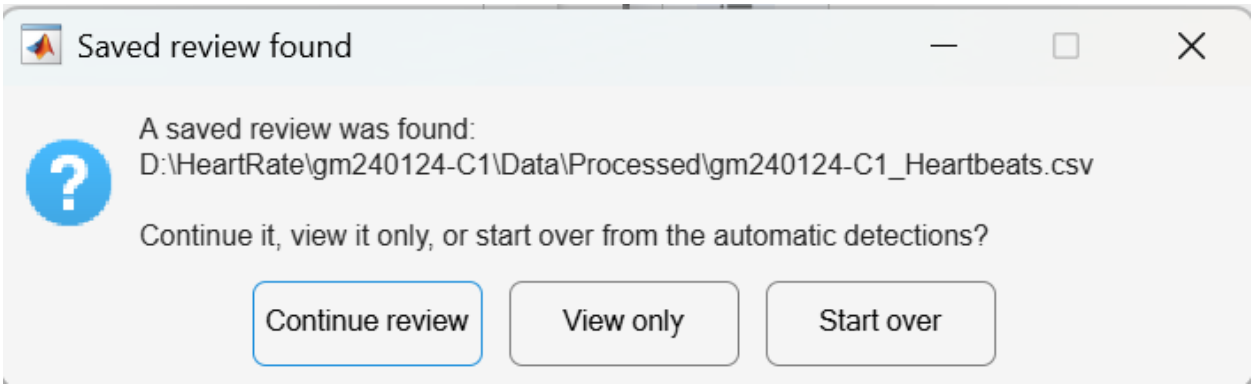
If any of the settings are different, you will see a popup that looks like this



If you rerun the script with different settings, you will get a popup box that look like this, explaining what is different. You have three options, to update the Settings sheet, to continue without updating the settings sheet, or to stop the run.

Note: when using different settings, you will override the “deployment_Settings” datasheet. I have not found this to be an issue. Most of these settings are just aids to get to the final manual review stage.

Once you run the Review section, you will get a popup box that looks like this



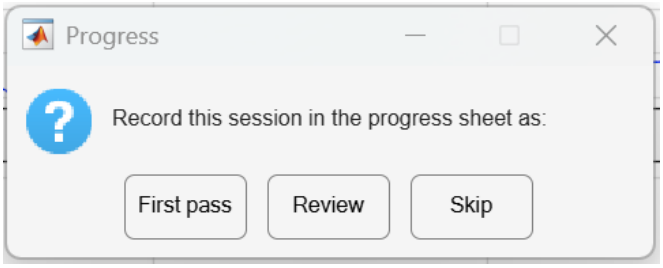
Continue review: reopens the saved review from “deployment_Heartbeats”, with all beats, x’s and comments exactly as they were saved, and all editing options available. The window opens at the start of the tag-on period; use the checkpoint time in “HeartRate_Progress” to find where work stopped.

View only: shows the saved review with editing switched off, so nothing can be changed by accident. Useful for checking someone else’s work. Close the window with the **Close** button.

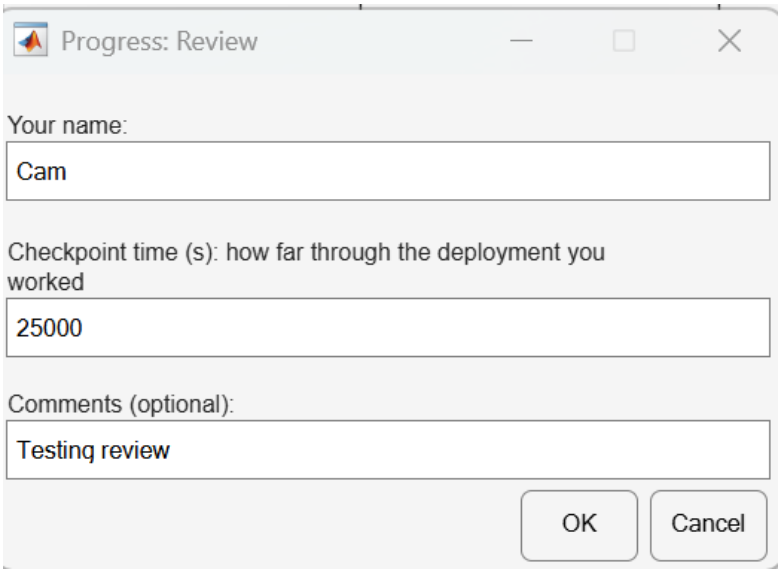
Start over: discards the saved review and starts again from the beats found by the automatic detector (with x’s pre-placed again). A confirmation popup appears first. The saved file is only replaced when you next save, so closing without saving leaves it unchanged; once saved, all previous edits and comments are lost.

When you are done reviewing for the session, whether you completed only a portion of the deployment or the entire thing, click “**Save & finish**”

A popup box will appear, click “**Review**”



The next popup box will appear. Enter your name, the time in the deployment that you have reviewed through, and any comments you want to make.



And one final box will appear. Click **"Yes"** if you have completed the deployment review and **"No"** if not.

At this stage, your "HeartRate_Progress" sheet should have the review information entered automatically. This will allow you to see which deployments have been completely reviewed.

